INCH-POUND

MIL-DTL-83723/81F
1 October 2010
SUPERSEDING
MIL-DTL-83723/81E
20 October 2000

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SINGLE HOLE MOUNT, BAYONET COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H, J, L, P AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83723.

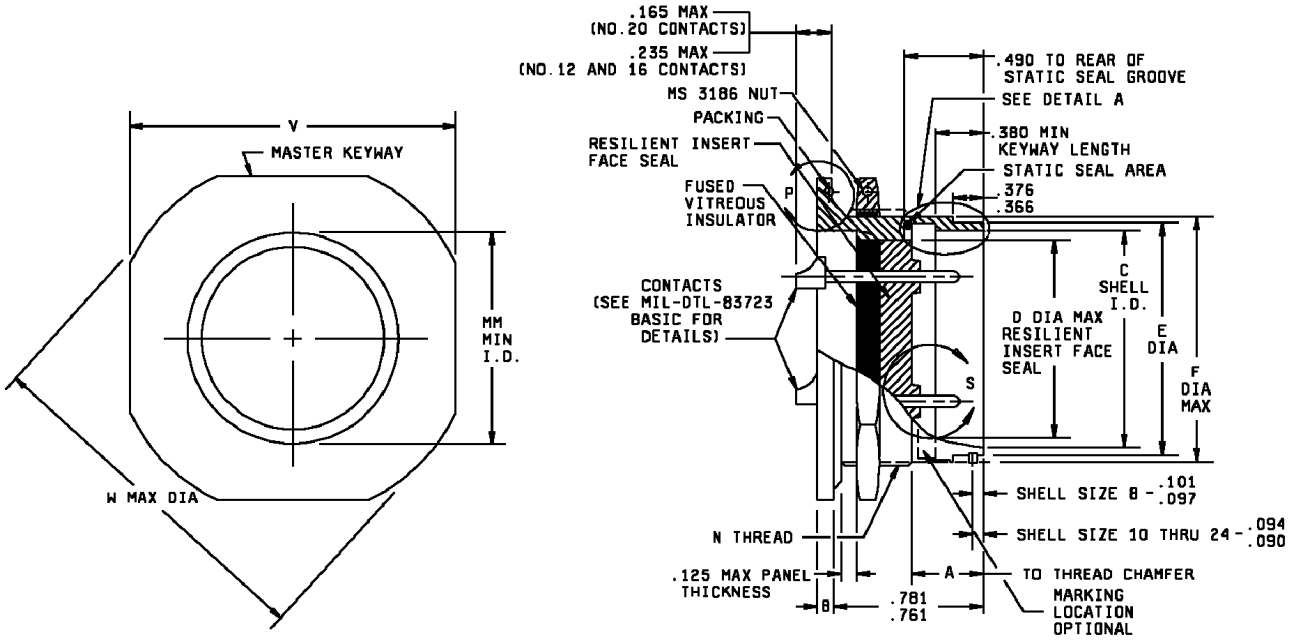


FIGURE 1. Dimensions and configurations.

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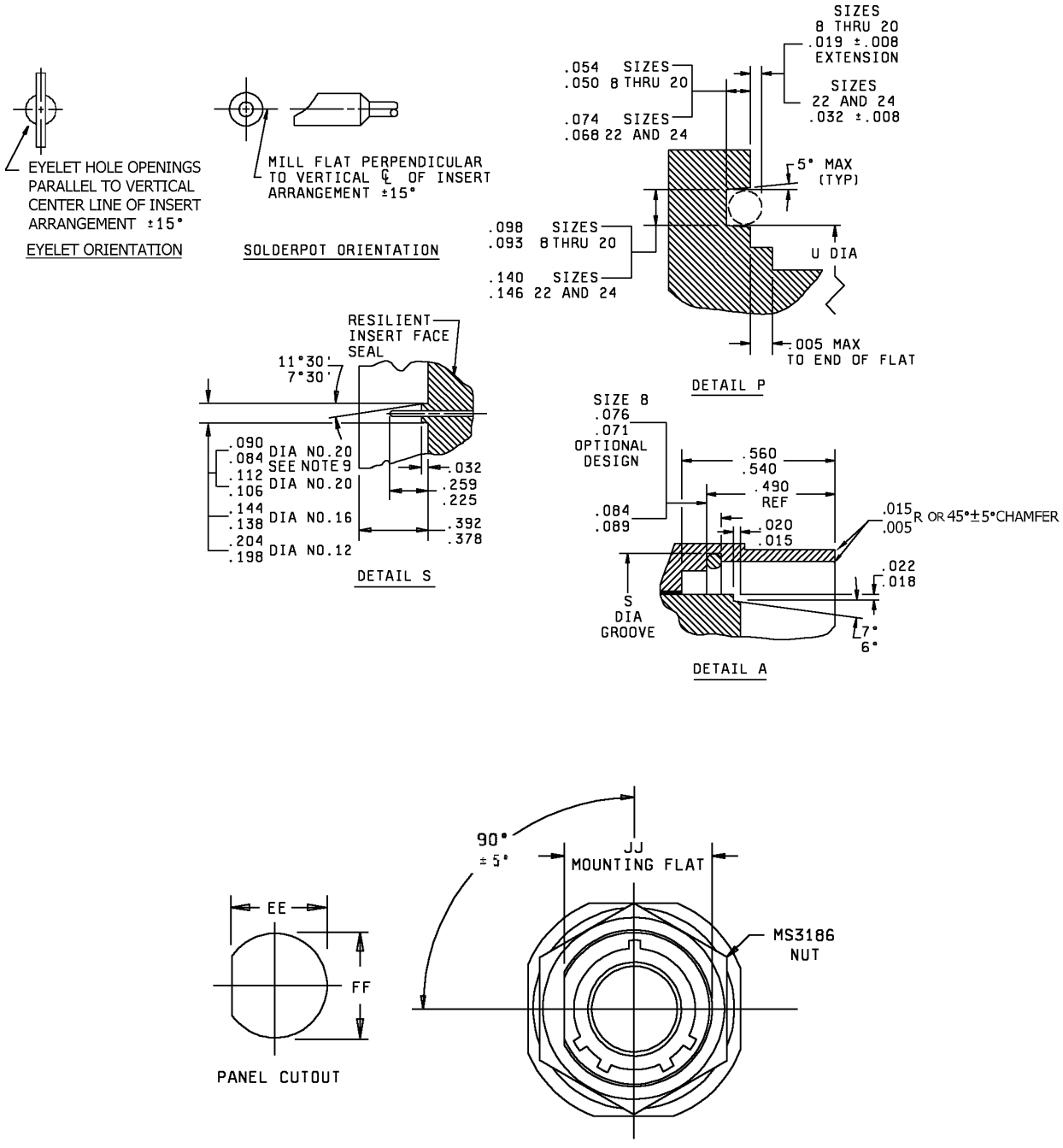


FIGURE 1. Dimensions and configurations - Continued.

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Shell size	A	B	C	D	E	F	N thread
8	.481 .461	.137 .097	.428 .433	.312	.536 .531	.561	.625-20 UN-2A
10	.481 .461	.137 .097	.530 .535	.388	.659 .654	.696	.750-20 UNEF-2A
12	.481 .461	.137 .097	.700 .705	.558	.829 .824	.875	.9375-20 UNEF-2A
14	.481 .461	.137 .097	.769 .774	.627	.898 .893	.935	1.000-20 UNEF-2A
16	.481 .461	.137 .097	.896 .901	.772	1.025 1.020	1.062	1.125-20 UN-2A
18	.481 .461	.137 .097	1.002 1.007	.860	1.131 1.126	1.187	1.250-18 UNEF-2A
20	.481 .461	.137 .097	1.127 1.132	.985	1.256 1.251	1.312	1.375-18 UNEF-2A
22	.481 .461	.168 .128	1.252 1.257	1.110	1.381 1.376	1.437	1.500-18 UNEF-2A
24	.481 .461	.168 .128	1.377 1.382	1.235	1.506 1.501	1.562	1.625-18 UNEF-2A

FIGURE 1. Dimensions and configurations - Continued.

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Shell size	U	V	W	S + .005 - .000	EE	FF	MM	JJ
8	.670	.984 .974	1.068	.508	.605	.635	.493	.596 .590
10	.777	1.109 1.099	1.192	.636	.730	.760	.555	.721 .715
12	.965	1.296 1.286	1.380	.806	.917	.947	.743	.908 .902
14	1.090	1.396 1.386	1.505	.875	.980	1.010	.805	.971 .965
16	1.215	1.521 1.511	1.630	1.002	1.105	1.135	.931	1.096 1.090
18	1.340	1.646 1.636	1.765	1.108	1.229	1.260	1.055	1.220 1.214
20	1.465	1.771 1.761	1.860	1.233	1.354	1.385	1.243	1.345 1.339
22	1.574	1.959 1.949	2.068	1.358	1.479	1.510	1.305	1.470 1.464
24	1.699	2.084 2.074	2.160	1.483	1.604	1.635	1.493	1.595 1.589

FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.312	7.92	.761	19.33	1.099	27.91	1.386	35.20
.004	0.10	.366	9.30	.769	19.53	1.105	28.07	1.396	35.46
.005	0.13	.376	9.55	.772	19.61	1.109	28.17	1.437	36.50
.008	0.20	.378	9.60	.774	19.66	1.110	28.19	1.464	37.19
.015	0.38	.380	9.65	.777	19.74	1.125	28.58	1.465	37.21
.018	0.46	.388	9.86	.781	19.84	1.126	28.60	1.470	37.34
.019	0.48	.392	9.96	.805	20.45	1.127	28.63	1.479	37.57
.020	0.51	.428	10.87	.824	20.93	1.131	28.73	1.493	37.92
.022	0.56	.433	11.00	.826	20.98	1.132	28.75	1.500	38.10
.032	0.81	.461	11.71	.829	21.06	1.135	28.82	1.501	38.13
.050	1.27	.481	12.21	.860	21.84	1.182	30.02	1.506	38.25
.054	1.37	.490	12.45	.875	22.23	1.187	30.15	1.510	38.35
.068	1.73	.493	12.52	.893	22.68	1.192	30.28	1.511	38.38
.071	1.80	.530	13.46	.896	22.76	1.214	30.84	1.521	38.63
.074	1.88	.531	13.49	.898	22.81	1.215	30.86	1.562	39.67
.076	1.93	.535	13.59	.901	22.89	1.220	30.99	1.574	39.98
.084	2.13	.536	13.61	.902	22.91	1.229	31.22	1.589	40.36
.089	2.26	.540	13.72	.908	23.06	1.235	31.37	1.595	40.51
.090	2.29	.555	14.10	.917	23.29	1.243	31.57	1.604	40.74
.093	2.36	.558	14.17	.931	23.65	1.250	31.75	1.625	41.28
.094	2.39	.560	14.22	.935	23.75	1.251	31.78	1.630	41.40
.097	2.46	.561	14.25	.9357	23.77	1.252	31.80	1.635	41.53
.098	2.49	.590	14.99	.942	23.93	1.256	31.90	1.636	41.55
.101	2.57	.596	15.14	.947	24.05	1.257	31.93	1.646	41.81
.106	2.69	.605	15.37	.965	24.51	1.260	32.00	1.699	43.15
.112	2.84	.625	15.88	.971	24.66	1.286	32.66	1.756	44.60
.125	3.18	.627	15.93	.974	24.74	1.296	32.92	1.771	44.98
.128	3.25	.635	16.13	.980	24.89	1.305	33.15	1.860	47.24
.137	3.48	.654	16.61	.984	24.99	1.312	33.32	1.949	49.50
.138	3.51	.659	16.74	.985	25.02	1.339	34.01	1.959	49.76
.140	3.58	.670	17.02	1.000	25.40	1.340	34.03	2.068	52.53
.144	3.66	.696	17.68	1.002	25.45	1.345	34.16	2.074	52.68
.146	3.71	.700	17.78	1.010	25.65	1.354	34.39	2.084	52.93
.165	4.19	.705	17.92	1.020	25.91	1.375	34.93	2.160	54.86
.168	4.27	.715	18.16	1.025	26.04	1.376	34.95		
.198	5.03	.721	18.31	1.055	26.80	1.377	34.98		
.204	5.18	.730	18.54	1.062	26.97	1.380	35.05		
.225	5.72	.743	18.87	1.068	27.13	1.381	35.08		
.235	5.97	.750	19.05	1.090	27.69	1.382	35.10		
.259	6.58	.760	19.30	1.096	27.84	1.385	35.18		

FIGURE 1. Dimensions and configurations - Continued.

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NOTES: (Notes 4 through 8 apply to contacts only).

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 inches (0.13 mm) for three place decimals, ± .01 inches (0.25 mm) for two place decimals and ± 1° for all angles.
4. Dimensions apply after plating.
5. Surface finish: 63 **(** $\sqrt{63}$ **)** all over in accordance with ASME-B46.1, unless otherwise specified.
6. Remove all burrs and sharp edges .003 to .005 inches (0.076 to 0.127 mm) R.
7. Diameters to be concentric within .004 inches (0.102 mm) TIR.
8. Diameter in the same cross-sectional plane to be concentric with each other within .003 inches (0.076 mm) TIR.
9. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

This connector mates with MIL-DTL-83723/75 and /77.

For insert arrangements, see MIL-STD-1554 and table I.

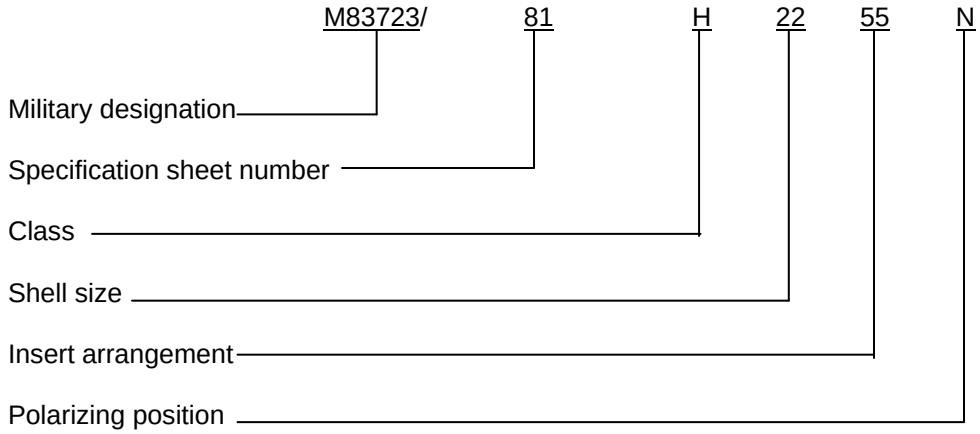
TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangements	Contact cavities
8	03, 98	1, 2, and 3
10	06	1, 2, 3, 4, 5, and 6

Connector accessories: See SAE-AS85049.

O-ring: See MIL-DTL-83723.

Part or Identifying Number (PIN) example:



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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83723, this document references the following:

MIL-DTL-83723/75
MIL-DTL-83723/77
MIL-STD-1554
MS3186
ASME-B46.1
SAE-AS85049

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force – 85
DLA - CC

Review activities:
Army - AR, AV, MI
Navy - EC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5935-2010-050)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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